

Everything you always wanted to know about XML parsing

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Thank You

- ASF
- xerces-j-dev
- xerces-j-user



Outline

- Basic XML Concepts
- SAX Parsing
- DOM Parsing
- JAXP
- Namespaces
- XML Schema



Basic XML Concepts

- Well formedness
- Validity
  - DTD's
  - Schemas
- Entities

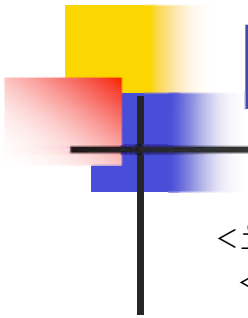
Example: RSS

```
<?xml version="1.0" encoding="ISO-8859-1"?>

<!DOCTYPE rss PUBLIC "-//Netscape Communications//DTD RSS .91//EN"
            "http://my.netscape.com/publish/formats/rss-0.91.dtd">

<rss version="0.91">

 <channel>
  <title>freshmeat.net</title>
  <link>http://freshmeat.net</link>
  <description>the one-stop-shop for all your Linux software
needs</description>
  <language>en</language>
  <rating>(PICS-1.1 "http://www.classify.org/safesurf/" 1 r (SS~~000
1))</rating>
  <copyright>Copyright 1999, Freshmeat.net</copyright>
  <pubDate>Thu, 23 Aug 1999 07:00:00 GMT</pubDate>
  <lastBuildDate>Thu, 23 Aug 1999 16:20:26 GMT</lastBuildDate>
  <docs>http://www.blahblah.org/fm.cdf</docs>
```



RSS 2

```
<image>
 <title>freshmeat.net</title>
 <url>http://freshmeat.net/images/fm.mini.jpg</url>
 <link>http://freshmeat.net</link>
 <width>88</width>
 <height>31</height>
 <description>This is the Freshmeat image stupid</description>
</image>

<item>
 <title>kdbg 1.0beta2</title>
 <link>http://www.freshmeat.net/news/1999/08/23/935449823.html</link>
 <description>This is the Freshmeat image stupid</description>
</item>

<item>
 <title>HTML-Tree 1.7</title>
 <link>http://www.freshmeat.net/news/1999/08/23/935449856.html</link>
 <description>This is the Freshmeat image stupid</description>
</item>
```

RSS 3

```
  <textinput>
   <title>quick finder</title>
   <description>Use the text input below to search freshmeat</description>
   <name>query</name>
   <link>http://core.freshmeat.net/search.php3</link>
  </textinput>

  <skipHours>
   <hour>2</hour>
  </skipHours>

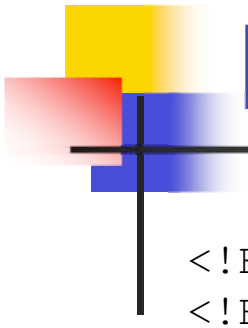
  <skipDays>
   <day>1</day>
  </skipDays>

 </channel>
</rss>
```

RSS DTD

```
<!ELEMENT rss (channel)>
<!ATTLIST rss
          version     CDATA #REQUIRED> <!-- must be "0.91"> -->

<!ELEMENT channel (title | description | link | language | item+ |
                   rating?| image? | textinput? | copyright? | pubDate? |
                   lastBuildDate? | docs? | managingEditor? |
                   webMaster? | skipHours? | skipDays?)*>
<!ELEMENT title (#PCDATA)>
<!ELEMENT description (#PCDATA)>
<!ELEMENT link (#PCDATA)>
<!ELEMENT language (#PCDATA)>
<!ELEMENT rating (#PCDATA)>
<!ELEMENT copyright (#PCDATA)>
<!ELEMENT pubDate (#PCDATA)>
<!ELEMENT lastBuildDate (#PCDATA)>
<!ELEMENT docs (#PCDATA)>
<!ELEMENT managingEditor (#PCDATA)>
<!ELEMENT webMaster (#PCDATA)>
<!ELEMENT hour (#PCDATA)>
<!ELEMENT day (#PCDATA)>
<!ELEMENT skipHours (hour+)>
<!ELEMENT skipDays (day+)>
```



RSS DTD 2

```
<!ELEMENT item (title | link | description)*>
<!ELEMENT textinput (title | description | name | link)*>
<!ELEMENT name (#PCDATA)>

<!ELEMENT image (title | url | link | width? | height? |
          description?)*>
<!ELEMENT url (#PCDATA)>
<!ELEMENT width (#PCDATA)>
<!ELEMENT height (#PCDATA)>
```

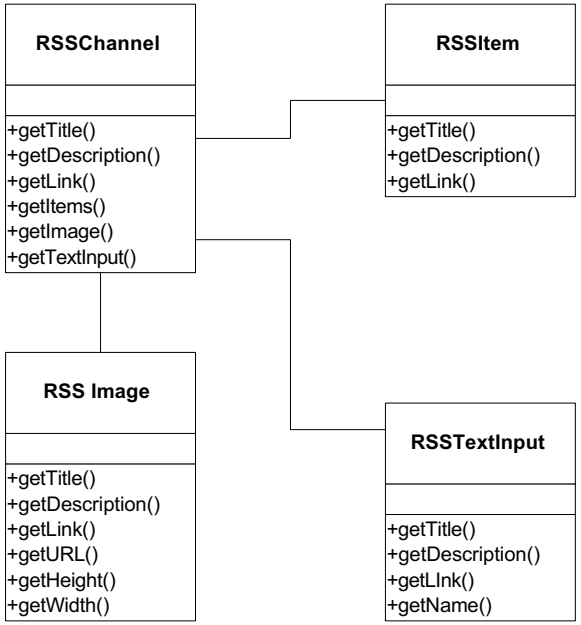


XML Application Tasks

- Convert XML into application object
- Convert application object to XML
- Read or write XML into a database
- XSLT conversion to XHTML or HTML

RSS Objects

Logical View



RSSItem 1

```
public class RSSItem extends Object implements java.io.Serializable {

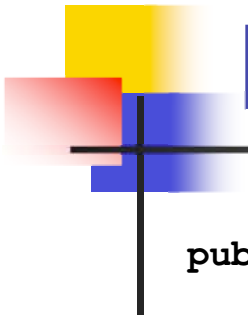
  private String title;
  private String description;
  private String link;

  public String getTitle () {
    return title;
  }

  public void setTitle (String value) {
    String oldValue = title;
    title = value;
  }

  public String getLink () {
    return link;
  }

  public void setLink (String value) {
    String oldValue = link;
    link = value;
  }
```



RSSItem 2

```
public String getDescription () {
    return description;
  }

  public void setDescription (String value) {
    String oldValue = description;
    description = value;
  }
}
```



Outline

- Basic XML Concepts
- SAX Parsing
- DOM Parsing
- JAXP
- Namespaces
- XML Schema



Parser API's

- What is the job of a parser API?
  - Make the structure and contents of a document available
  - Report errors



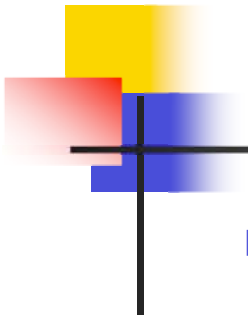
SAX API

- Event callback style
- Representation of elements & atts
  - must build own stack
- SAX Processing Pipeline model
- Development model
  - Reasonably open
  - xml-dev
  - http://www.saxproject.org



ContentHandler

- The basic SAX2 handler for XML
- Callbacks for
  - Start of document
  - End of document
  - Start of element
  - End of element
  - Character data
  - Ignorable whitespace



Handler Strategies

- Single Monolithic
  - Requires the handler to know entirely about the grammar
- One per application object and multiplexer
  - Each application object has its own ContentHandler
  - The Multiplexer handler is registered with the parser
  - Multiplexer is responsible for swapping SAX handlers as the context changes.

SAX 2 RSS Handler

```
public void startElement(String namespaceURI, String localName,
                         String qName, Attributes atts) throws SAXException {
  currentText = new StringBuffer();
  textStack.push(currentText);
  if (localName.equals("rss")) {
    elementStack.push(localName);
    if (!atts.getValue("version").equals("0.91"))
      throw new SAXException("Incorrect RSS Version.");
  } else if (localName.equals("channel")) {
    elementStack.push(localName);
    currentChannel = new RSSChannel();
    currentItems = new Vector();
    currentChannel.setItems(currentItems);
    currentImage = new RSSImage();
    currentChannel.setImage(currentImage);
    currentTextInput = new RSSTextInput();
    currentChannel.setTextInput(currentTextInput);
    currentChannel.setSkipHours(new Vector());
    currentChannel.setSkipDays(new Vector());
  } else if (localName.equals("title")) {
  } else if (localName.equals("description")) {
  } else if (localName.equals("link")) {
  } else if (localName.equals("language")) {
```

SAX 2 RSS Handler 2

```
    } else if (localName.equals("item")) {
      elementStack.push(localName);
      currentItem = new RSSItem();
    } else if (localName.equals("rating")) {
    } else if (localName.equals("image")) {
      elementStack.push(localName);
    } else if (localName.equals("textinput")) {
      elementStack.push(localName);
    } else if (localName.equals("copyright")) {
    } else if (localName.equals("pubDate")) {
    } else if (localName.equals("lastBuildDate")) {
    } else if (localName.equals("docs")) {
    } else if (localName.equals("managingEditor")) {
    } else if (localName.equals("webMaster")) {
    } else if (localName.equals("hour")) {
    } else if (localName.equals("day")) {
    } else if (localName.equals("skipHours")) {
      elementStack.push(localName);
    } else if (localName.equals("skipDays")) {
      elementStack.push(localName);
    } else {
    }
  }
```

SAX 2 RSS Handler 3

```
public void endElement(String namespaceURI, String localName, String qName)
            throws SAXException {
  try {
    String stackValue = (String) elementStack.peek();
    String text = ((StringBuffer)textStack.pop()).toString();
    if (localName.equals("rss")) {
      elementStack.pop();
    } else if (localName.equals("channel")) {
      String e = (String) elementStack.pop();
    } else if (localName.equals("title")) {
      if (stackValue.equals("channel")) {
        currentChannel.setTitle(text);
      } else if (stackValue.equals("image")) {
        currentImage.setTitle(text);
      } else if (stackValue.equals("item")) {
        currentItem.setTitle(text);
      } else if (stackValue.equals("textinput")) {
        currentTextInput.setTitle(text);
      } else {
      }
```

SAX 2 RSS Handler 4

```
} else if (localName.equals("description")) {
  if (stackValue.equals("channel")) {
    currentChannel.setDescription(text);
  } else if (stackValue.equals("image")) {
    currentImage.setDescription(text);
  } else if (stackValue.equals("item")) {
    currentItem.setDescription(text);
  } else if (stackValue.equals("textinput")) {
    currentTextInput.setDescription(text);
  } else {
  }
} else if (localName.equals("link")) {
  if (stackValue.equals("channel")) {
    currentChannel.setLink(text);
  } else if (stackValue.equals("image")) {
    currentImage.setLink(text);
  } else if (stackValue.equals("item")) {
    currentItem.setLink(text);
  } else if (stackValue.equals("textinput")) {
    currentTextInput.setLink(text);
  } else {
  }
```

SAX 2 RSS Handler 5

```
} else if (localName.equals("language")) {
  currentChannel.setLanguage(text);
} else if (localName.equals("item")) {
  currentItems.add(currentItem);
  String e = (String) elementStack.pop();
} else if (localName.equals("rating")) {
  currentChannel.setRating(text);
} else if (localName.equals("image")) {
  currentChannel.setImage(currentImage);
  String e = (String) elementStack.pop();
} else if (localName.equals("height")) {
  currentImage.setHeight(text);
} else if (localName.equals("width")) {
  currentImage.setWidth(text);
} else if (localName.equals("url")) {
  currentImage.setURL(text);
} else if (localName.equals("textinput")) {
  currentChannel.setTextInput(currentTextInput);
  String e = (String) elementStack.pop();
} else if (localName.equals("name")) {
  currentTextInput.setName(text);
```

SAX 2 RSS Handler 6

```
    } else if (localName.equals("copyright")) {
      currentChannel.setCopyright(text);
    } else if (localName.equals("pubDate")) {
      currentChannel.setPubDate(text);
    } else if (localName.equals("lastBuildDate")) {
      currentChannel.setLastBuildDate(text);
    } else if (localName.equals("docs")) {
      currentChannel.setDocs(text);
    } else if (localName.equals("managingEditor")) {
      currentChannel.setManagingEditor(text);
    } else if (localName.equals("webMaster")) {
      currentChannel.setWebMaster(text);
    } else if (localName.equals("hour")) {
      currentHour = text;
    } else if (localName.equals("day")) {
      currentDay = text;
    } else if (localName.equals("skipHours")) {
      currentChannel.getSkipHours().add(currentHour);
      String e = (String) elementStack.pop();
    } else if (localName.equals("skipDays")) {
      currentChannel.getSkipDays().add(currentDay);
      String e = (String) elementStack.pop();
    } else {}
  } catch (Exception e) {
    e.printStackTrace();
  }
```

SAX 2 RSS Handler 7

```
public void characters(char[] ch,int start,int length) throws SAXException {
  currentText.append(ch, start, length);
}
```



SAX 2 Basic Event Handling

- Basic Event Handling
  - Start tags in startElement
  - End tags in endElement
  - attribute handling in startElement
  - characters() for content
  - character data handling in endElement
    - Multiple characters calls

SAX 2 RSS ErrorHandler

```
public void warning(SAXParseException ex) throws SAXException {
  System.err.println("[Warning] "+
  getLocationString(ex)+": "+
  ex.getMessage());
}

public void error(SAXParseException ex) throws SAXException {
  System.err.println("[Error] "+
  getLocationString(ex)+": "+
  ex.getMessage());
}

public void fatalError(SAXParseException ex) throws SAXException {
  System.err.println("[Fatal Error] "+
  getLocationString(ex)+": "+
  ex.getMessage());
  throw ex;
}
```

SAX 2 RSS ErrorHandler 2

```
private String getLocationString(SAXParseException ex) {
  StringBuffer str = new StringBuffer();

  String systemId = ex.getSystemId();
  if (systemId != null) {
    int index = systemId.lastIndexOf('/');
    if (index != -1)
      systemId = systemId.substring(index + 1);
    str.append(systemId);
  }
  str.append(':');
  str.append(ex.getLineNumber());
  str.append(':');
  str.append(ex.getColumnNumber());

  return str.toString();
}
```



SAX 2 Error Handling

- Error Handling
  - warning
  - error – XML 1.0 spec errors
    - parser may recover
  - fatalError – XML 1.0 spec errors

SAX 2 RSS Driver

```
public static void main(String args[]) {
  XMLReader r = new SAXParser();
  ContentHandler h = new SAX2RSSHandler();
  r.setContentHandler(h);
  ErrorHandler eh = new SAX2RSSErrorHandler();
  r.setErrorHandler(eh);
  try {
    r.parse("file:///Work/workspace-1.4/apachecon2002/fm0.91_full.rdf");
  } catch (SAXException se) {
    System.out.println("SAX Error during parsing "+se.getMessage());
    se.printStackTrace();
  } catch (IOException e) {
    System.out.println("I/O Error during parsing "+e.getMessage());
    e.printStackTrace();
  } catch (Exception e) {
    System.out.println("Error during parsing "+e.getMessage());
    e.printStackTrace();
  }
  System.out.println(((SAX2RSSHandler) h).getChannel().toString());
}
```



Entity Handling

- Entity handling
  - Pluggable entity handling is important for server applications.
    - Provides access point to DBMS, LDAP, etc.

Entity Handler

- `<!DOCTYPE rss PUBLIC "-//Netscape Communications//DTD RSS91//EN" "http://my.netscape.com/publish/formats/rss-0.91.dtd">`
- This will do a network fetch!
- To avoid the network fetch:

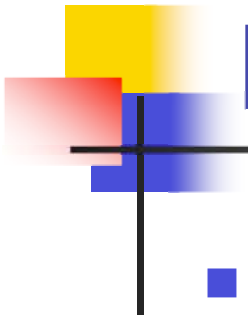
```
public class SAX2RSSEntityResolver implements EntityResolver {

  public SAX2RSSEntityHandler() {}

  public InputSource resolveEntity(String publicId,String systemId)
    throws SAXException, IOException {
    if (publicId.equals("-//Netscape Communications//DTD RSS 0.91//EN")) {
      FileReader r =
        new FileReader("/Work/workspace-1.4/apachecon2002/rss-0.91.dtd");
      return new InputSource(r);
    }
    return null;
  }
}
```

EntityDriver

```
public static void main(String[] args) {
  XMLReader r = new SAXParser();
  ContentHandler h = new SAX2RSSHandler();
  r.setContentHandler(h);
  ErrorHandler eh = new SAX2RSSErrorHandler();
  r.setErrorHandler(eh);
  EntityResolver er = new SAX2RSSEntityResolver();
  r.setEntityResolver(er);
  try {
    r.parse("file:///Work/workspace-1.4/apachecon2002/fm0.91_full.rdf");
  } catch (SAXException se) {
    System.out.println("SAX Error " + se.getMessage());
    se.printStackTrace();
  } catch (IOException e) {
    System.out.println("I/O Error " + e.getMessage());
    e.printStackTrace();
  } catch (Exception e) {
    System.out.println("Error " + e.getMessage());
    e.printStackTrace();
  }
  System.out.println(((SAX2RSSHandler) h).getChannel().toString());
}
```



DefaultHandler

- The kitchen sink
- Lets you do ContentHandler, ErrorHandler, EntityResolver and DTD Handler
- A good way to go if you are only overriding a small number of methods



SAX 2 Extension Handlers

- LexicalHandler
  - Entity boundaries
  - DTD bounaries
  - CDATA sections
  - Comments
- DeclHandler
  - ElementDecls and AttributeDecls
  - Internal and External Entity Decls
  - NO parameter entities



SAX 2 Configuration

- Uses strings (URI's) as lookup keys
- Features - booleans
  - validation
  - namespaces
- Properties - Objects
  - Extensibility by returning an object that implements additional function

Configuration Example

```
public static void main(String[] argv) {
  // construct parser; set features
  XMLReader parser = new SAXParser();
  try {
    parser.setFeature("http://xml.org/sax/features/namespaces", true);
    parser.setFeature("http://xml.org/sax/features/validation", true);
    parser.setProperty(
      "http://xml.org/sax/properties/declaration-handler",dh);
  } catch (SAXException e) {
    e.printStackTrace(System.err);
  }
}
```



Xerces configuration

- Standard way to set all parser configuration settings.
- Applies to SAX and DOM Parsers
  - XNI Parsers too



Xerces Features

- Inclusion of general entities
- Inclusion of parameter entities
- Dynamic validation
- Extra warnings
- Allow use of java encoding names
- Lazy evaluating DOM
- DOM EntityReference creation
- Inclusion of igorable whitespace in DOM
- Schema validation [also full checking]
- Control of Non-Validating behavior
  - Defaulting attributes



Xerces Properties

- Name of DOM implementation class
- DOM Element node currently being parsed
- Values for
  - noNamespaceSchemaLocation
  - SchemaLocation
- Parser input buffer size



Validation

- When to use validation
  - System boundaries
- non-validating != wf checking
  - Not required to read external entities
- validation dial settings



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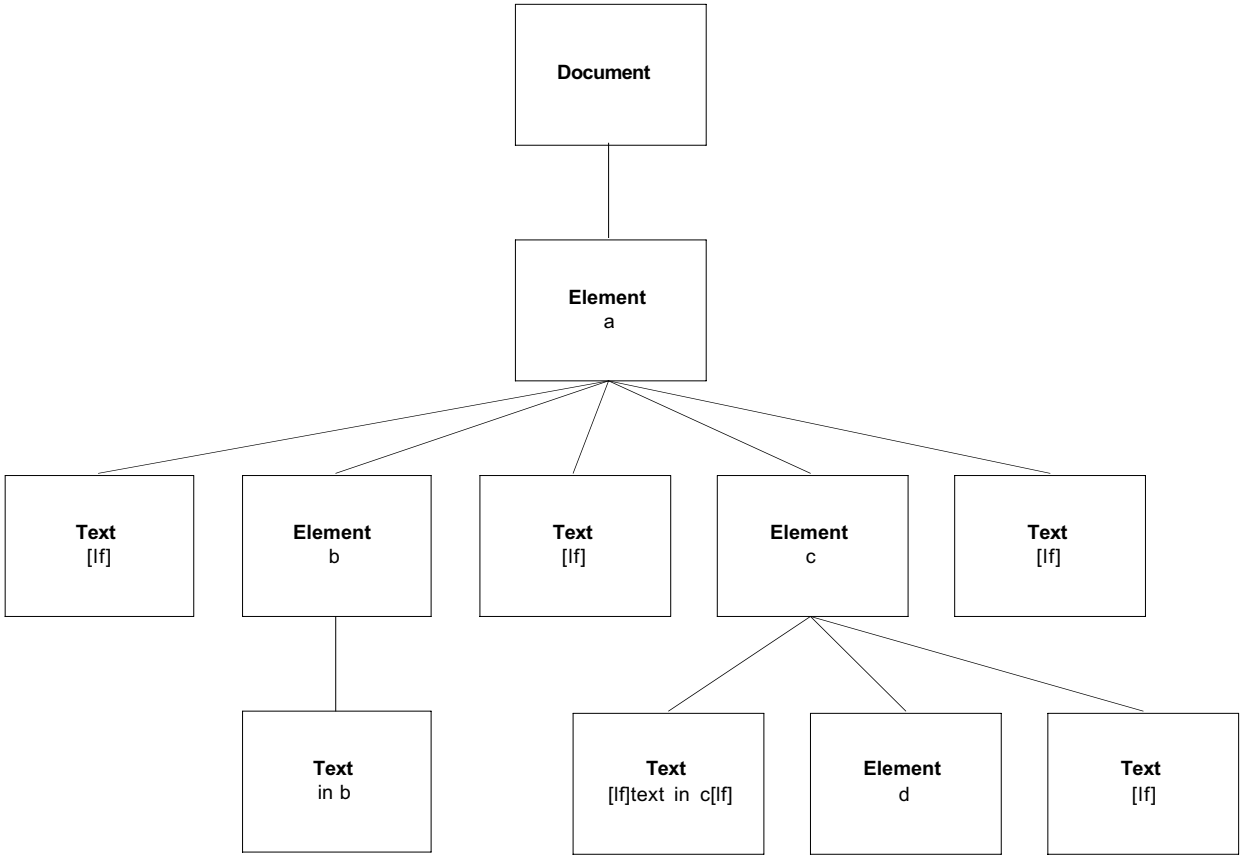
DOM API

- Object representation of elements and attributes
- Development Model
  - W3C Recommendations and process
- http:///www.w3.org/DOM

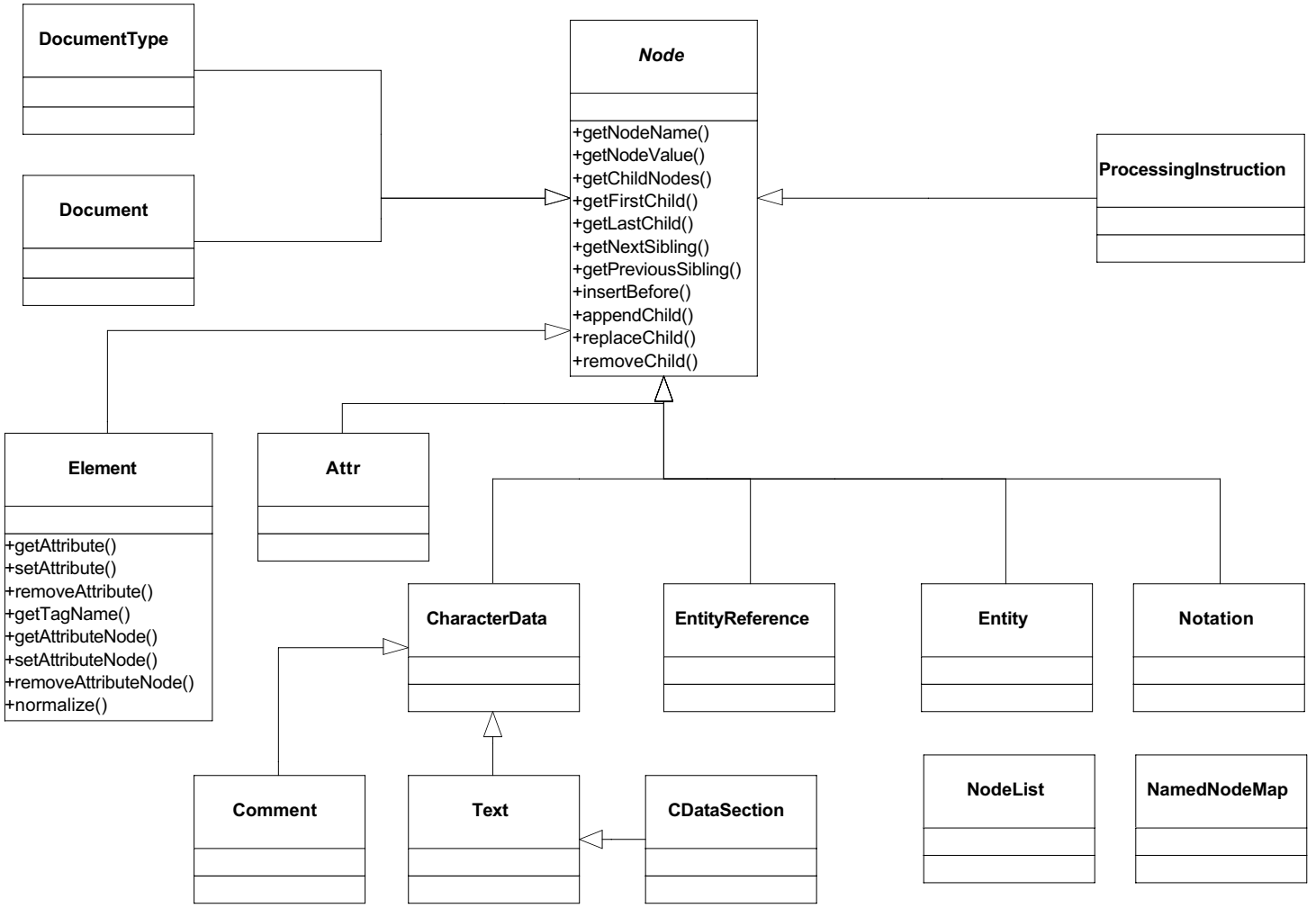
DOM Instance (XML)

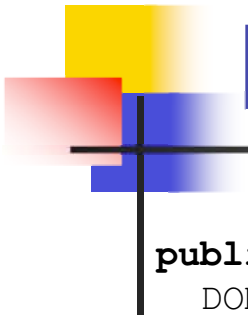
```
<?xml version="1.0" encoding="US-ASCII"?>
<a>
 <b>in b</b>
 <c>
  text in c
  <d/>
 </c>
</a>
```

DOM Tree


Document
Element
a
Text
[lf]
Element
b
Text
[lf]
Element
c
Text
[lf]
Text
in b
Text
[lf]text in c[lf]
Element
d
Text
[lf]

DOM Architecture





DOM RSS

```
public static void main(String args[]) {
  DOMParser p = new DOMParser();
  try {
    p.parse("file:///work/workspace-1.4/apachecon2002/fm0.91_full.rdf");
    Document d = p.getDocument();
    System.out.println(dom2Channel(d).toString());

  } catch (SAXException se) {
    System.out.println("Error during parsing "+se.getMessage());
    se.printStackTrace();
  } catch (IOException e) {
    System.out.println("I/O Error during parsing "+e.getMessage());
    e.printStackTrace();
  }
}
```

DOM2Channel

```
public static RSSChannel dom2Channel(Document d) {
  RSSChannel c = new RSSChannel();
  NodeList nl = d.getElementsByTagName("rss");
  Element rss = (Element) nl.item(0);
  if (!rss.getAttribute("version").equals("0.91")) return null;

  nl = d.getElementsByTagName("channel");
  Element channel = (Element) nl.item(0);
  for (Node n = channel.getFirstChild();
       n != null;
       n = n.getNextSibling()) {
    if (n.getNodeType() != Node.ELEMENT_NODE) continue;
    Element e = (Element) n;
    e.normalize();
    String text = e.getFirstChild().getNodeValue();
    if (e.getTagName().equals("title")) {
      c.setTitle(text);
    } else if (e.getTagName().equals("description")) {
      c.setDescription(text);
    } else if (e.getTagName().equals("link")) {
      c.setLink(text);
    } else if (e.getTagName().equals("language")) {
...
```

DOM2Channel 2

…

```
    }
    nl = channel.getElementsByTagName("item");
    c.setItems(dom2Items(nl));
    nl = channel.getElementsByTagName("image");
    c.setImage(dom2Image(nl.item(0)));
    nl = channel.getElementsByTagName("textinput");
    c.setTextInput(dom2TextInput(nl.item(0)));
    nl = channel.getElementsByTagName("skipHours");
    c.setSkipHours(dom2SkipHours(nl.item(0)));
    nl = channel.getElementsByTagName("skipDays");
    c.setSkipDays(dom2SkipDays(nl.item(0)));
    return c;
}
```



DOM Node Traversal Methods

- Two paradigms
- Directly from the Node
  - `Node#getFirstChild`
  - `Node#getNextSibling`
- From the Nodelist for a Node
  - `Node#getChildren`
  - `NodeList#getLength()`
  - `NodeList#item(int i)`

DOM2Items

```
public static Vector dom2Items(NodeList nl) {
  Vector result = new Vector();
  Node item = null;
  for (int x = 0; x < nl.getLength(); x++) {
    item = nl.item(x);
    if (item.getNodeType() != Node.ELEMENT_NODE) continue;
    RSSItem i = new RSSItem();
    for (Node n = item.getFirstChild(); n != null; n = n.getNextSibling()) {
      if (n.getNodeType() != Node.ELEMENT_NODE) continue;
      Element e = (Element) n;
      e.normalize();
      String text = e.getFirstChild().getNodeValue();
      if (e.getTagName().equals("title")) {
        i.setTitle(text);
      } else if (e.getTagName().equals("description")) {
        i.setDescription(text);
      } else if (e.getTagName().equals("link")) {
        i.setLink(text);
      }
    }
    result.add(i);
  }
  return result;
}
```



DOM Level 1

- Attr
  - API for result as strings
- No DTD support
- Need Import - copying between DOMs is disallowed



DOM Level 2

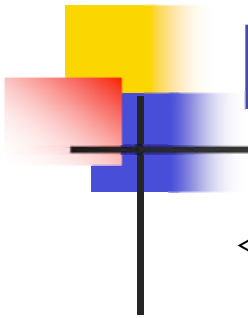
- Core / Namespaces
  - Import
- Traversal
  - Views on DOM Trees
- Events
- Range
- Stylesheets

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Xerces DOM extensions

- Feature controls insertion of ignorable whitespace
- Feature controls creation of EntityReference Nodes
- User data object provided on implementation class org.apache.xml.dom.NodeImpl

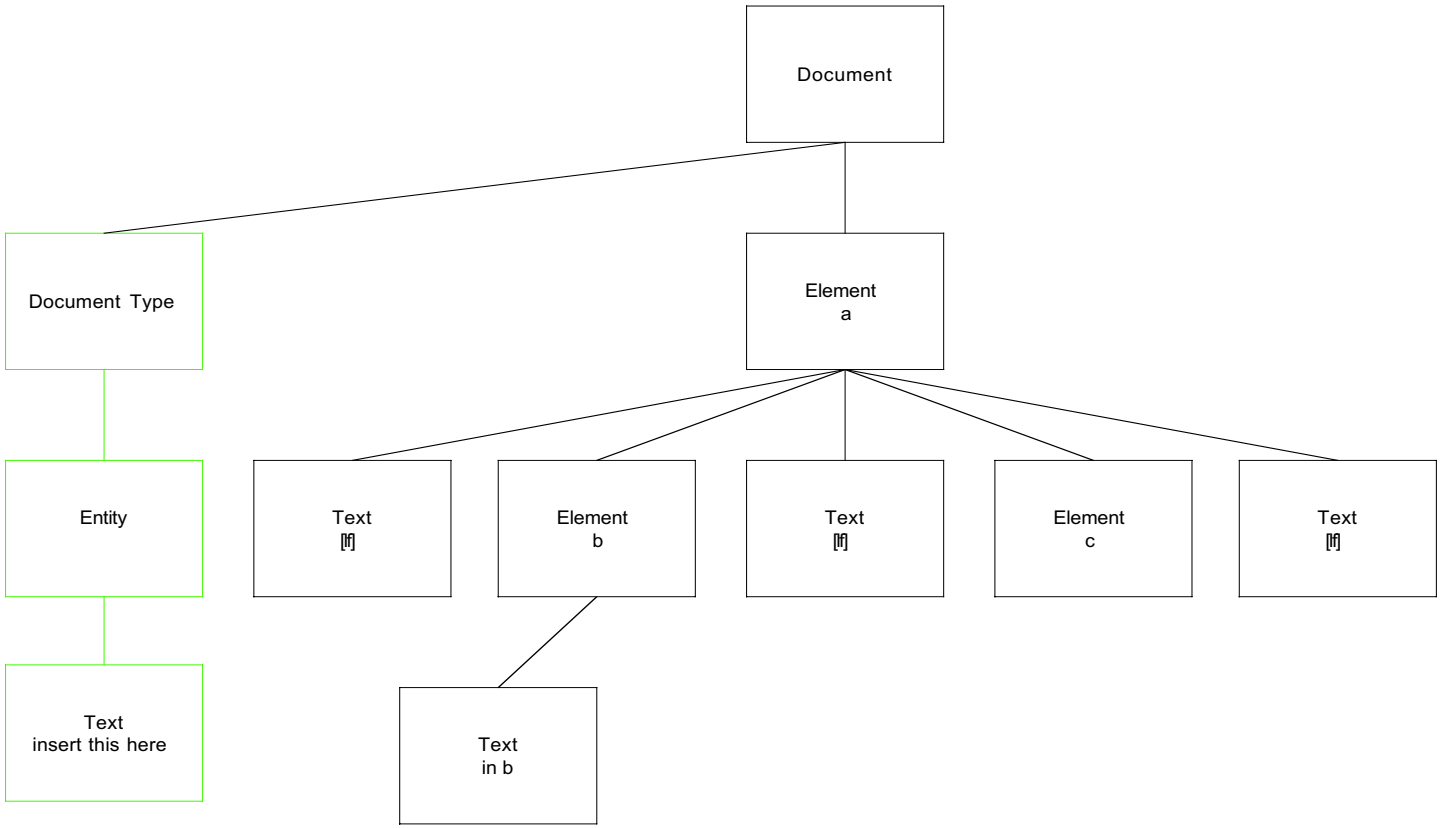


DOM Entity Reference Nodes 1

```
<?xml version="1.0" encoding="US-ASCII"?>
<!DOCTYPE a [
<!ENTITY boilerplate "insert this here">
]>
<a>
 <b>in b</b>
 <c>
  text in c but &boilerplate;
  <d/>
 </c>
</a>
```

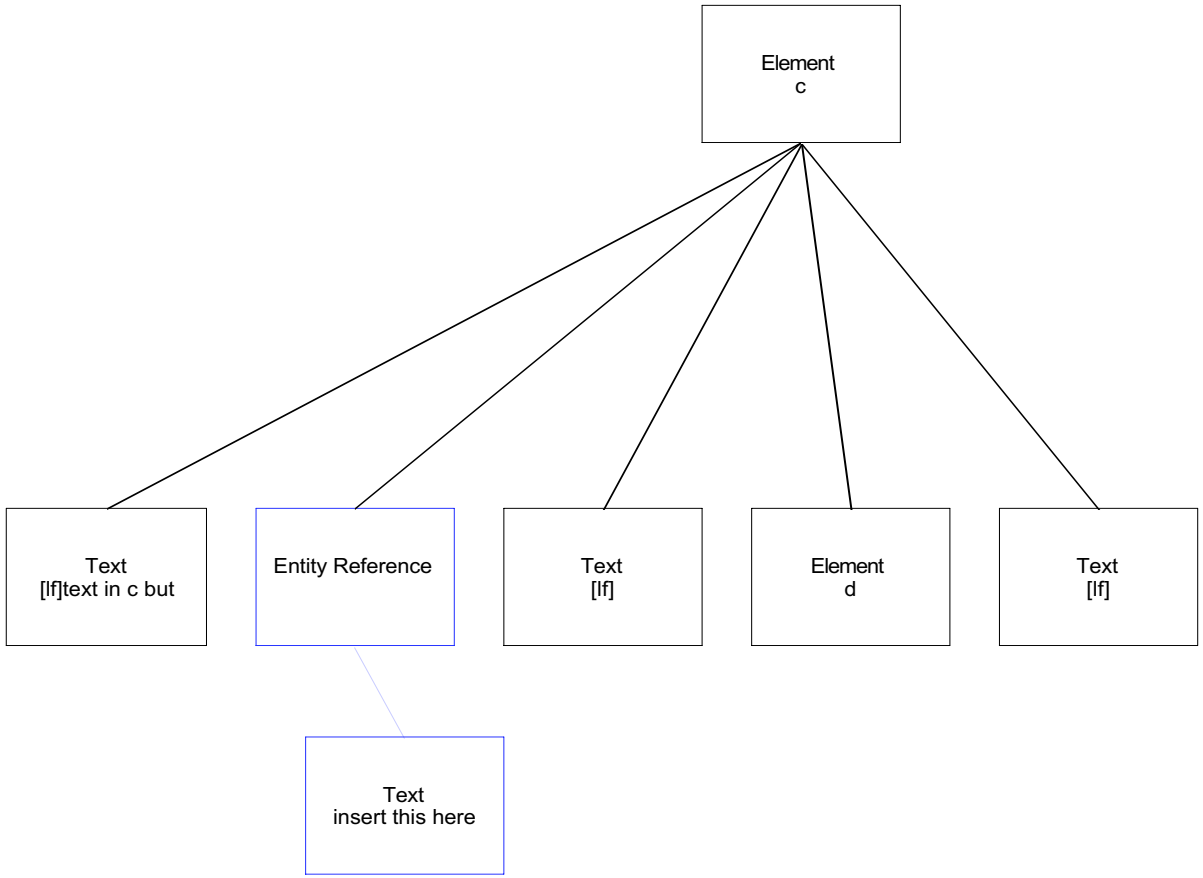
DOM Entity Reference Nodes 2

- Document Type Node



DOM Entity Reference Nodes 3

- Entity Reference Node





Xerces Lazy/Deferred DOM

- Delays object creation
  - a node is accessed - its object is created
  - all siblings are also created.
- Reduces time to complete parsing
- Reduces memory usage
- Good if you only access part of the DOM
- Bad on small documents



Tricky DOM stuff

- Entity and EntityReference nodes – depends on parser settings
- Ignorable whitespace
- Subclassing the DOM
  - Use the Document factory class



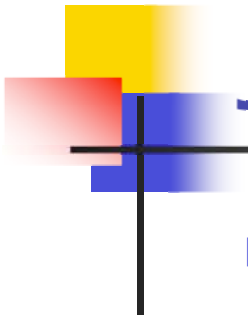
DOM vs SAX

- DOM
  - creates a document object tree
  - tree must the be reparsed & converted to BO's
  - Could subclass BO's from DOM classes
- SAX
  - event handler based API
  - stack management
  - no intermediate data structures generated



Outline

- Basic XML Concepts
- SAX Parsing
- DOM Parsing
- JAXP
- Namespaces
- XML Schema



JAXP

- Java API for XML Parsing
- Sun JSR-063
- Version 1.1
- Abstractions for SAX, DOM, XSLT
- Support in Xerces

SAX in JAXP

```
public static void main(String args[]) {
  XMLReader r = null;

  SAXParserFactory spf = SAXParserFactory.newInstance();
  spf.setNamespaceAware(true);

  try {
    r = spf.newSAXParser().getXMLReader();
  } catch (Exception e) {
    System.exit(1);
  }

  ContentHandler h = new SAX2RSSHandler();
  r.setContentHandler(h);
  ErrorHandler eh = new SAX2RSSErrorHandler();
  r.setErrorHandler(eh);
  try {
    r.parse("file:///work/workspace-1.4/apachecon2002/fm0.91_full.rdf");
  } catch (Exception e) {
    System.out.println("Error during parsing "+e.getMessage());
    e.printStackTrace();
  }
  System.out.println(((SAX2RSSHandler) h).getChannel().toString());
}
```

DOM in JAXP

```
public static void main(String args[]) {
  DocumentBuilderFactory dbf = DocumentBuilderFactory.newInstance();
  dbf.setNamespaceAware(true);
  dbf.setValidating(true);

  DocumentBuilder db = null;

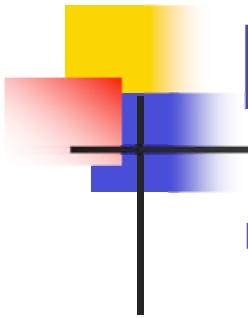
  try {
    db = dbf.newDocumentBuilder();
  } catch (Exception e) {
    System.exit(1);
  }

  try {
    Document doc = db.parse(
      "file:///work/workspace-1.4/apachecon2002/fm0.91_full.rdf");
  } catch (SAXException se) {
  } catch (IOException ioe) {
  }
  System.out.println(dom2Channel(doc).toString());
}
```



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Namespaces

- Purpose
  - Syntactic discrimination only
  - They don't point to anything
  - They don't imply schema composition/combination
- Universal Names
  - URI + Local Name
    {http://www.mydomain.com}Element
- Declarations
  - xmlns "attribute"
- Prefixes
- Scoping
- Validation and DTDs

Namespace Example

```
<xsl:stylesheet version="0.0"
                xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
                xmlns:html="http://www.w3.org/TR/xhtml1/strict">
 <xsl:template match="/">
   <html:html>
     <html:head>
       <html:title>Title</html:title>
    </html:head>
    <html:body>
      <xsl:apply-templates/>
    </html:body>
 </xsl:template>

 ...

</xsl:stylesheet>
```

Default Namespace Example

```
<xsl:stylesheet version="1.0"
                xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
                xmlns="http://www.w3.org/TR/xhtml1/strict">
 <xsl:template match="/">
   <html>
     <head>
       <title>Title</title>
     </head>
     <body>
       <xsl:apply-templates/>
     </body>
 </xsl:template>

 ...

</xsl:stylesheet>
```



Outline

- Basic XML Concepts
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XML Schema

- Richer grammar specification
- W3C Recommendation May 2001
- Structures
  - XML Instance Syntax
  - Namespaces
  - Rich content models
- Datatypes
  - Lots
- Support in Xerces 1.4+

RSS Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE xs:schema
          PUBLIC "-//W3C//DTD XMLSCHEMA 200102//EN"
          "XMLSchema.dtd" >

 <xs:schema
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://my.netscape.com/publish/formats/rss-0.91"
  xmlns="http://my.netscape.com/publish/formats/rss-0.91">
  <xs:element name="rss">
   <xs:complexType>
    <xs:sequence>
     <xs:element ref="channel"/>
    </xs:sequence>
    <xs:attribute name="version" type="xs:string" fixed="0.91"/>
   </xs:complexType>
  </xs:element>
```

RSS Schema 2

```
<xs:element name="channel">
   <xs:complexType>
    <xs:choice minOccurs="0" maxOccurs="unbounded“>
      <xs:element ref="title"/>
      <xs:element ref="description"/>
      <xs:element ref="link"/>
      <xs:element ref="language"/>
      <xs:element ref="item" minOccurs="1" maxOccurs="unbounded"/>
      <xs:element ref="rating" minOccurs="0"/>
      <xs:element ref="image" minOccurs="0"/>
      <xs:element ref="textinput" minOccurs="0"/>
      <xs:element ref="copyright" minOccurs="0"/>
      <xs:element ref="pubDate" minOccurs="0"/>
      <xs:element ref="lastBuildDate" minOccurs="0"/>
      <xs:element ref="docs" minOccurs="0"/>
      <xs:element ref="managingEditor" minOccurs="0"/>
      <xs:element ref="webMaster" minOccurs="0"/>
      <xs:element ref="skipHours" minOccurs="0"/>
      <xs:element ref="skipDays" minOccurs="0"/>
     </xs:choice>
   </xs:complexType>
  </xs:element>
```

RSS Schema 3

```
…
  <xs:element name="url"  type="xs:anyURI"/>
  <xs:element name="name" type="xs:string"/>
  <xs:element name="rating" type="xs:string"/>
  <xs:element name="language" type="xs:string"/>
  <xs:element name="width" type="xs:integer"/>
  <xs:element name="height" type="xs:integer"/>
  <xs:element name="copyright" type="xs:string"/>
  <xs:element name="pubDate" type="xs:string"/>
  <xs:element name="lastBuildDate" type="xs:string"/>
  <xs:element name="docs" type="xs:string"/>
  <xs:element name="managingEditor" type="xs:string"/>
  <xs:element name="webMaster" type="xs:string"/>
  <xs:element name="hour" type="xs:integer"/>
  <xs:element name="day" type="xs:integer"/>
</xs:schema>
```



Schema Processing in Xerces

- Features needed for validation
  - http://xml.org/sax/features/validation
  - http://xml.org/sax/features/namespaces
  - http://apache.org/xml/features/validation/schema
- Other features
  - http://apache.org/xml/features/validation/schema-full-checking

Processing Schemas

```
public static void main(String[] args) {
  SAXParserFactory spf = SAXParserFactory.newInstance();
  spf.setNamespaceAware(true);

  XMLReader r = null;

  try {
    r = spf.newSAXParser().getXMLReader();
  } catch (Exception e) {
    System.exit(1);
  }

  ContentHandler h = new SAX2RSSHandler();
  r.setContentHandler(h);
  ErrorHandler eh = new SAX2RSSErrorHandler();
  r.setErrorHandler(eh);

  // turn on schema validation
  try {
    r.setFeature("http://xml.org/sax/features/validation", true);
    r.setFeature("http://xml.org/sax/features/namespaces", true);
    r.setFeature("http://apache.org/xml/features/validation/schema", true);
  } catch (Exception e) {
  }

  try {
    r.parse("file:///work/workspace-1.4/apachecon2002/fm0.91_full-schema.rdf");
  } catch (Exception e) {
    e.printStackTrace();
  }
  System.out.println(((SAX2RSSHandler) h).getChannel().toString());
}
```



Finding the schema document

- Other properties
  - http://apache.org/xml/properties/schema/external-schemaLocation
  - http://apache.org/xml/properties/schema/external-noNamespaceSchemaLocation



Thank You!

- http://xml.apache.org
- twl@apache.org
- http://www.sauria.com/blog

This presenation has ended.
Click anywhere to return.

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